

JavaScript Notes — Day 4

Chapter 4: Arrays

What is an Array?

An array is a collection of items, stored in a single variable and accessed by index (starting at 0).

```
let marks = [12, 44, 98, 56];
console.log(marks);

// access by index
marks[2]; // 98

// update an element
marks[2] = 33;
```

Looping Over an Array

Using a for loop (with index)

```
let heroes = ["ironman", "thor", "spiderman", "hulk"];
for (let ind = 0; ind < heroes.length; ind++) {
  console.log(heroes[ind]);
}
```

Using a for...of loop (with value directly)

```
for (let hero of heroes) {
  console.log(hero);
}
```

Practice Problems

1. Find the average of the array values

```
let marks = [85, 34, 66, 78, 98];
let sum = 0;

for (let mark of marks) {
  sum += mark / marks.length;
}
console.log(sum); // average
```

2. Apply a discount to each value in an array

```
let items = [250, 645, 300, 900, 50];
let i = 0;

for (let val of items) {
```

```
let off = val / 10;
items[i] -= off;
console.log("value after offer:", items[i]);
i++;
}
```

Array Methods

Some methods change the original array, and some return a new array without touching the original.

Method	Purpose	Effect on Original Array
push()	Adds element(s) to the end	Changes original array
pop()	Deletes last element & returns it	Changes original array
unshift()	Adds element(s) to the start	Changes original array
shift()	Deletes first element & returns it	Changes original array
splice()	Add, remove, or replace elements	Changes original array
toString()	Converts array to a comma-separated string	No change — returns new string
concat()	Joins multiple arrays	No change — returns new array
slice(start, end)	Extracts a portion of the array	No change — returns new array

Note: push(), pop(), unshift(), and shift() are the ones that return a deleted/added value where relevant (pop and shift return the removed element).

push() and concat()

```
let player = ["virat", "ronaldo", "rohit", "messi", "neymar"];
player.push("mbappe"); // adds to end, changes player

let hero = ["batman", "superman", "aquaman"];
let dep = player.concat(hero); // new array, player unchanged
console.log(dep);
```

pop() and toString()

```
player.pop(); // removes "mbappe", changes player
console.log(player.toString()); // "virat,ronaldo,rohit,messi,neymar"
```

unshift()

```
player.unshift("king"); // adds "king" to the start
```

Note: unshift(king) needs king as a string — "king" — otherwise JavaScript looks for a variable named king.

shift()

```
let hero = ["hjb", "fgy", "uhg"];
let val = hero.shift(); // removes "hjb", returns it
console.log(val); // "hjb"
```

slice(start, end)

Extracts elements from start up to (but not including) end. Original array is unchanged.

```
let player = ["virat", "ronaldo", "rohit", "messi", "neymar"];
console.log(player.slice(1, 5)); // ["ronaldo","rohit","messi","neymar"]
```

splice(startIndex, deleteCount, newElements...)

Changes the original array directly — can add, remove, or replace elements in place.

```
let arr = [1, 2, 3, 4, 5, 6, 7];

arr.splice(2, 2, 101, 102); // replace 2 elements at index 2 with 101, 102
console.log(arr); // [1, 2, 101, 102, 5, 6, 7]

arr.splice(2, 0, 101); // add element (deleteCount = 0)
arr.splice(3, 1); // delete element (no new elements passed)
arr.splice(2, 1, 5); // replace 1 element with 5
```

Example walkthrough: `[1,2,3,4,5,6,7].splice(2,2,101,102)` removes the 2 elements starting at index 2 (the 3 and 4) and inserts 101, 102 in their place → `[1,2,101,102,5,6,7]`.